LIMITS OF FINITE NONEMPTY SETS OVER COFILTERED DIAGRAMS ARE NONEMPTY

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3087-3120, following the statement at 3079-3085.

Editorial note. The proof is the classical Zorn/compactness argument; no external citations beyond the in-chapter reduction lemma.

Allowed standard prerequisites (editorial).

- Cofiltered diagrams reduce to directed inverse systems. Lemma lemma-directed-category-system of this chapter, invoked at proof lines 3088-3089 (carded as L26113). Identifies the two statements of the lemma, reducing to directed inverse systems over a directed set I.
- Zorn's lemma. Standard set-theoretic principle, applied at proof line 3102. The poset of subsystems ordered by inclusion has minimal elements; the chain lower bound uses finiteness of the S_i.

Source reference. Original lemma 0032: sources/stacks-categories/pinned-original-source.tex, source lines 2927–2927. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *If $S : \mathcal{I} \to Sets$ is a cofiltered diagram of sets and all the S_i are finite nonempty, then $\lim_i S_i$ is nonempty. In other words, the limit of a directed inverse system of finite nonempty sets is nonempty.*

Proof. The two statements are equivalent by Lemma 0032. Let I be a directed set and let $(S_i)_{i \in I}$ be an inverse system of finite nonempty sets over I. Let us say that a *subsystem* T is a family $T = (T_i)_{i \in I}$ of nonempty subsets $T_i \subset S_i$ such that $T_{i'}$ is mapped into T_i by the transition map $S_{i'} \to S_i$ for all $i' \geq i$. Denote $\mathcal{T}$ the set of subsystems. We order $\mathcal{T}$ by inclusion. Suppose T_α, $\alpha \in A$ is a totally ordered family of elements of $\mathcal{T}$. Say $T_\alpha = (T_{\alpha,i})_{i \in I}$. Then we can find a lower bound $T = (T_i)_{i \in I}$ by setting $T_i = \bigcap_{\alpha \in A} T_{\alpha,i}$ which is manifestly a finite nonempty subset of S_i as all the $T_{\alpha,i}$ are nonempty and as the T_α form a totally ordered family. Thus we may apply Zorn's lemma to see that $\mathcal{T}$ has minimal elements.

Let's analyze what a minimal element $T \in \mathcal{T}$ looks like. First observe that the maps $T_{i'} \to T_i$ are all surjective. Namely, as I is a directed set and T_i is finite, the intersection $T_i' = \bigcap_{i' \geq i} \mathrm{Im}(T_{i'} \to T_i)$ is nonempty. Thus $T' = (T_i')$ is a subsystem contained in T and by minimality $T' = T$. Finally, we claim that T_i is a singleton for each i. Namely, if $x \in T_i$, then we can define $T_{i'}' = (T_{i'} \to T_i)^{-1}(\{x\})$ for $i' \geq i$ and $T_j' = T_j$ if $j \not\geq i$. This is another subsystem as we've seen above that

Date: stacks-project:categories.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

the transition maps of the subsystem T are surjective. By minimality we see that $T = T'$ which indeed implies that T_i is a singleton. This holds for every $i \in I$, hence we see that $T_i = \{x_i\}$ for some $x_i \in S_i$ with $x_{i'} \mapsto x_i$ under the map $S_{i'} \to S_i$ for every $i' \geq i$. In other words, $(x_i) \in \lim S_i$ and the lemma is proved. $\square$